DATA ENGINEERING ASSIGNMENT

Apache Pig: Online Shopping Product Review Analysis

Filtering invalid ratings and calculating average rating by product category

Tool: Apache Pig (Pig Latin)

Storage: HDFS (Hadoop Distributed File System)

Dataset: Online shopping product reviews (CSV)

Execution mode shown in screenshots: Hadoop MapReduce

1. Aim

To use Apache Pig to process an online-shopping product-review dataset stored in HDFS, remove the CSV header and invalid ratings, group valid reviews by product category, and calculate the average rating for each category.

2. Problem Statement

An e-commerce company receives product reviews across categories such as Electronics, Home Appliances, Fashion, Books, and Beauty. Some review ratings may be missing or invalid. Create a Pig Latin script that loads the review data from HDFS, removes null ratings and ratings below 1.0, groups records by product category, and computes the average rating per category.

3. Software and Tools Used

- Apache Hadoop and HDFS — store and access the CSV input and output files.
- Apache Pig — express data processing using Pig Latin.
- Pig Latin — script language used for LOAD, FILTER, FOREACH, GROUP, AVG, and STORE operations.
- Linux terminal and nano editor — create the CSV file and Pig script and run commands.

4. Dataset: Online Shopping Product Reviews

The dataset contains 15 product reviews. The rating column is stored as text initially so that the script can explicitly exclude the literal value 'null' before converting valid ratings to numbers.

review_id	product_category	product_name	rating
R501	Electronics	Wireless_Earbuds	4.8
R502	Electronics	Bluetooth_Speaker	4.3
R503	Electronics	Smartwatch	0.5
R504	Home_Appliances	Air_Fryer	4.6
R505	Home_Appliances	Robot_Vacuum	4.2
R506	Home_Appliances	Electric_Kettle	null
R507	Fashion	Running_Shoes	4.5
R508	Fashion	Denim_Jacket	4.0
R509	Fashion	Backpack	0.0
R510	Books	Data_Science_Book	4.7
R511	Books	Programming_Guide	4.1
R512	Books	Machine_Learning_Book	4.4
R513	Beauty	Face_Cream	3.9
R514	Beauty	Shampoo	4.3
R515	Beauty	Hair_Serum	4.6

Invalid values included for testing: R503 has rating 0.5, R509 has rating 0.0, and R506 has rating null. All three should be excluded from the average calculations.

5. Step-by-Step Execution Commands

Step 1 — Start Hadoop services

```
start-dfs.sh
start-yarn.sh
jps
```

The start scripts start HDFS and YARN services. The jps command lists Java processes, which can help confirm that Hadoop daemons such as NameNode, DataNode, ResourceManager, and NodeManager are running. The exact processes shown depend on the installation.

[SCREENSHOT PLACEHOLDER 1: Paste the terminal screenshot showing start-dfs.sh, start-yarn.sh, and jps here.]

Step 2 — Create and inspect the CSV dataset

```
nano ~/shopping_reviews.csv
cat ~/shopping_reviews.csv
```

nano opens a text editor to create or edit the CSV file. The cat command prints the file contents so the dataset can be checked before uploading it to HDFS.

[SCREENSHOT PLACEHOLDER 2: Paste the screenshot showing shopping_reviews.csv and its rows here.]

Step 3 — Create an HDFS directory and upload the CSV

```
hdfs dfs -mkdir -p /user/aiml/shopping_reviews
hdfs dfs -put -f ~/shopping_reviews.csv /user/aiml/shopping_reviews/
```

mkdir -p creates the HDFS directory if it does not already exist. The -put command copies the local CSV file into HDFS; -f allows an existing destination file to be overwritten.

Step 4 — Verify the uploaded file

```
hdfs dfs -ls /user/aiml/shopping_reviews
hdfs dfs -cat /user/aiml/shopping_reviews/shopping_reviews.csv
```

The -ls command lists files and directories in HDFS. The -cat command displays the uploaded CSV contents, confirming that the file is available at the expected HDFS path.

[SCREENSHOT PLACEHOLDER 3: Paste the screenshot showing HDFS listing and hdfs dfs -cat output here.]

6. Apache Pig Script

Create the script file with:

```
nano ~/shopping_reviews_analysis.pig
```

Use the following Pig Latin code:

```
reviews = LOAD '/user/aiml/shopping_reviews/shopping_reviews.csv'
USING PigStorage(',')
AS (
    review_id:chararray,
    product_category:chararray,
    product_name:chararray,
    rating:chararray
);
```

```
data = FILTER reviews BY review_id != 'review_id';

non_null = FILTER data
BY rating IS NOT NULL AND rating != 'null';

typed_data = FOREACH non_null GENERATE
    review_id,
    product_category,
    product_name,
    (float)rating AS rating;

valid_reviews = FILTER typed_data BY rating >= 1.0;

grouped_reviews = GROUP valid_reviews BY product_category;

average_ratings = FOREACH grouped_reviews GENERATE
    group AS product_category,
    AVG(valid_reviews.rating) AS average_rating;

STORE average_ratings
INTO '/user/aiml/shopping_reviews/output'
USING PigStorage(',');
```

7. Explanation of the Pig Script

- **LOAD —** Reads the CSV file from the HDFS path.
- **PigStorage(',') —** Tells Pig that commas separate the fields in the CSV.
- **AS (...) —** Defines the four columns and their initial data types. rating is chararray because the CSV contains the text 'null'.
- **FILTER review_id != 'review_id' —** Removes the header row so the column names are not treated as a review.
- **FILTER rating IS NOT NULL AND rating != 'null' —** Removes missing ratings, including the literal text null in the CSV.
- **FOREACH ... GENERATE —** Creates a new version of each remaining row and casts rating to float so it can be used in arithmetic.
- **FILTER rating >= 1.0 —** Keeps ratings of 1.0 or above and removes 0.5 and 0.0.
- **GROUP ... BY product_category —** Places reviews with the same product category into groups.
- **AVG(valid_reviews.rating) —** Calculates the mean rating inside each category group.
- **STORE ... INTO ... —** Writes the category averages to the HDFS output directory.

8. Run the Script and Check the Output

First remove the old output directory if it exists, because Pig generally cannot write to an existing output directory:

```
hdfs dfs -rm -r -f /user/aiml/shopping_reviews/output
```

Run the Pig script in MapReduce mode:

```
pig -x mapreduce ~/shopping_reviews_analysis.pig
```

Pig parses the script, builds an execution plan, and submits the processing job to Hadoop. Log lines and warnings may appear in the terminal. The important check is whether the job completes and the output directory is created.

[SCREENSHOT PLACEHOLDER 4: Paste the terminal screenshot showing the Pig command and execution logs here.]

List the output directory and display the result:

```
hdfs dfs -ls /user/aiml/shopping_reviews/output
hdfs dfs -cat /user/aiml/shopping_reviews/output/part-r-00000
```

The output directory normally contains a part file and a _SUCCESS marker. The part file contains the category averages. The output order can vary because distributed processing does not guarantee a particular category order.

[SCREENSHOT PLACEHOLDER 5: Paste the terminal screenshot showing the output directory and the final average-rating results here.]

9. Expected Output and Calculation

The script starts with 15 review records. It removes one null rating and two ratings below 1.0, leaving 12 valid reviews. It then calculates one average for each of the five categories.

Product category	Valid ratings	Average rating (approx.)
Books	4.7, 4.1, 4.4	4.40
Beauty	3.9, 4.3, 4.6	4.2667
Electronics	4.8, 4.3	4.55
Fashion	4.5, 4.0	4.25
Home_Appliances	4.6, 4.2	4.40

The terminal may display more decimal places than the rounded values shown above. For example, the average for Beauty may appear as approximately 4.2666667 and Electronics as approximately 4.5500002 due to floating-point representation.

10. Screenshot Placement Checklist

The report includes explicit screenshot placeholders at the relevant steps. Paste the matching terminal screenshots at these positions if you want to replace the placeholders with images. Suggested mapping:

1. Screenshot 1: Hadoop startup — start-dfs.sh, start-yarn.sh, and jps.
2. Screenshot 2: Local CSV creation and dataset contents — nano/cat shopping_reviews.csv.
3. Screenshot 3: HDFS directory, uploaded file, and file verification.
4. Screenshot 4: Pig script execution command and execution logs.
5. Screenshot 5: HDFS output listing and final category averages.

11. Discussion Questions

What is Apache Pig?

Apache Pig is a high-level data-processing platform for Hadoop. It uses Pig Latin to describe data transformations.

What is HDFS?

HDFS stands for Hadoop Distributed File System. It stores files across a Hadoop cluster and provides access to those files.

What is Pig Latin?

Pig Latin is the scripting language used to express operations such as LOAD, FILTER, GROUP, FOREACH, and STORE.

Why is FILTER used?

FILTER keeps only records that satisfy a condition, such as removing null ratings or ratings below 1.0.

Why convert rating to float?

An average requires numeric values. Casting the rating to float allows Pig to calculate AVG.

Why GROUP BY product_category?

It collects reviews of the same category together so that a separate average can be calculated for each category.

Why remove the header?

The first CSV row contains column names, not a real customer review.

Why use Pig instead of writing raw MapReduce?

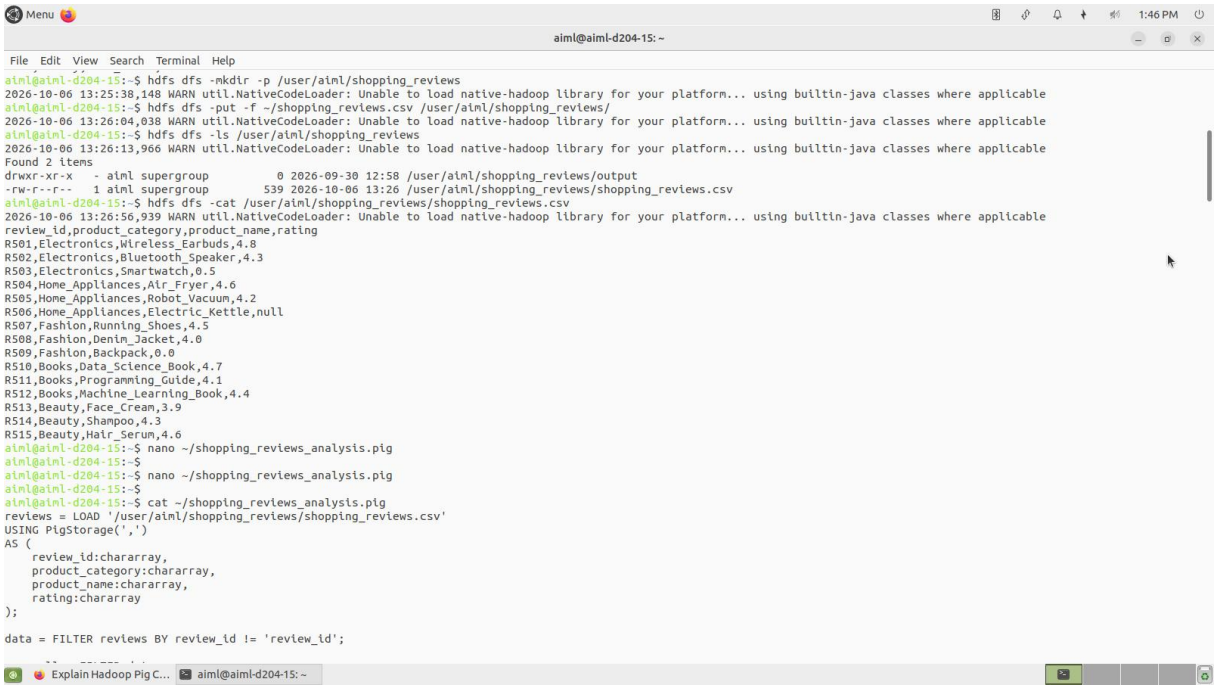
Pig Latin is shorter and higher-level for common data-flow operations. Pig translates the script into an execution plan for Hadoop.

12. Conclusion

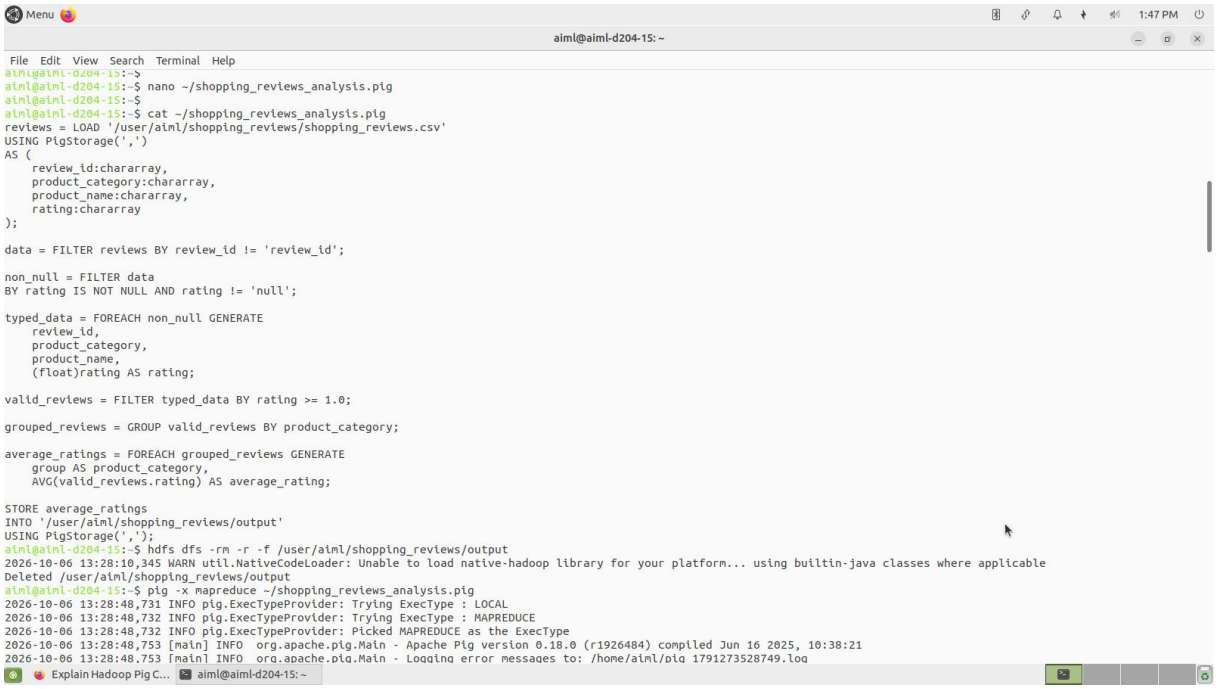
This assignment demonstrates an end-to-end Apache Pig workflow for online-shopping product reviews. The dataset is stored in HDFS, loaded by Pig, cleaned by removing the header and invalid ratings, converted to numeric ratings, grouped by product category, and aggregated using AVG. The results show the average rating for Books, Beauty, Electronics, Fashion, and Home Appliances.

Appendix A — Supplied Terminal Screenshots

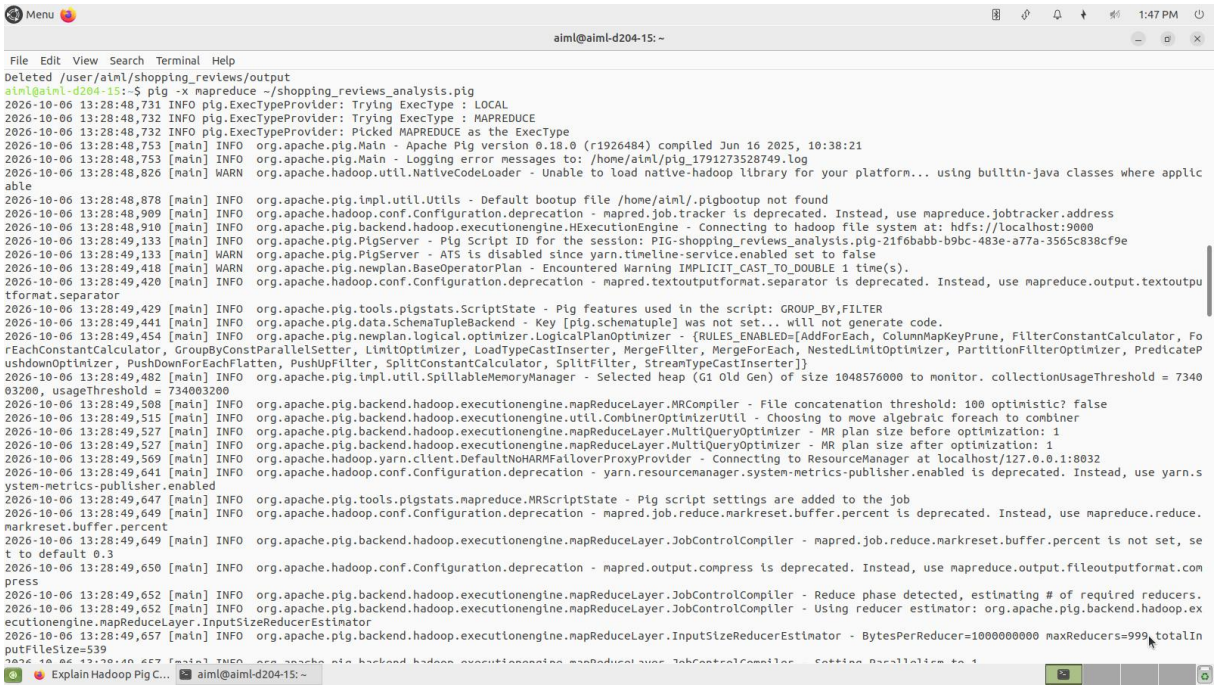
The following screenshots were supplied with the request. They are included here as reference evidence. Some screenshots show overlapping terminal content; use the screenshot-placement checklist above when choosing the clearest image for each placeholder.



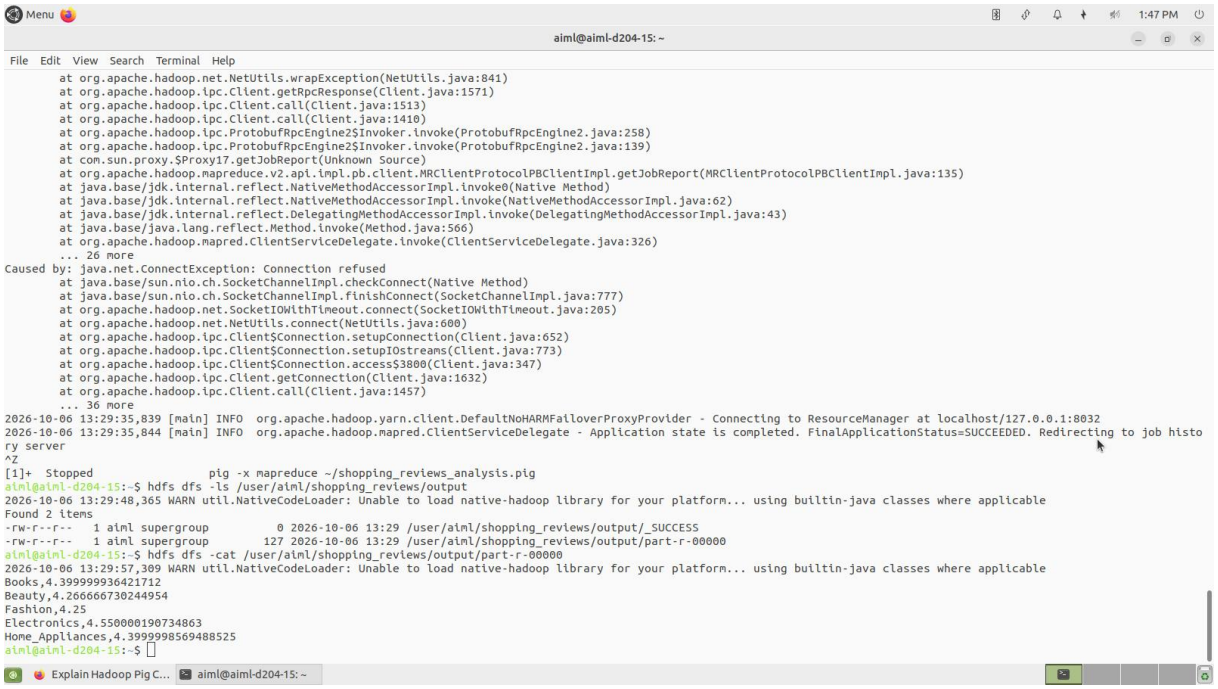
Screenshot 1 — HDFS setup, CSV upload, dataset verification, and Pig script contents (first part).



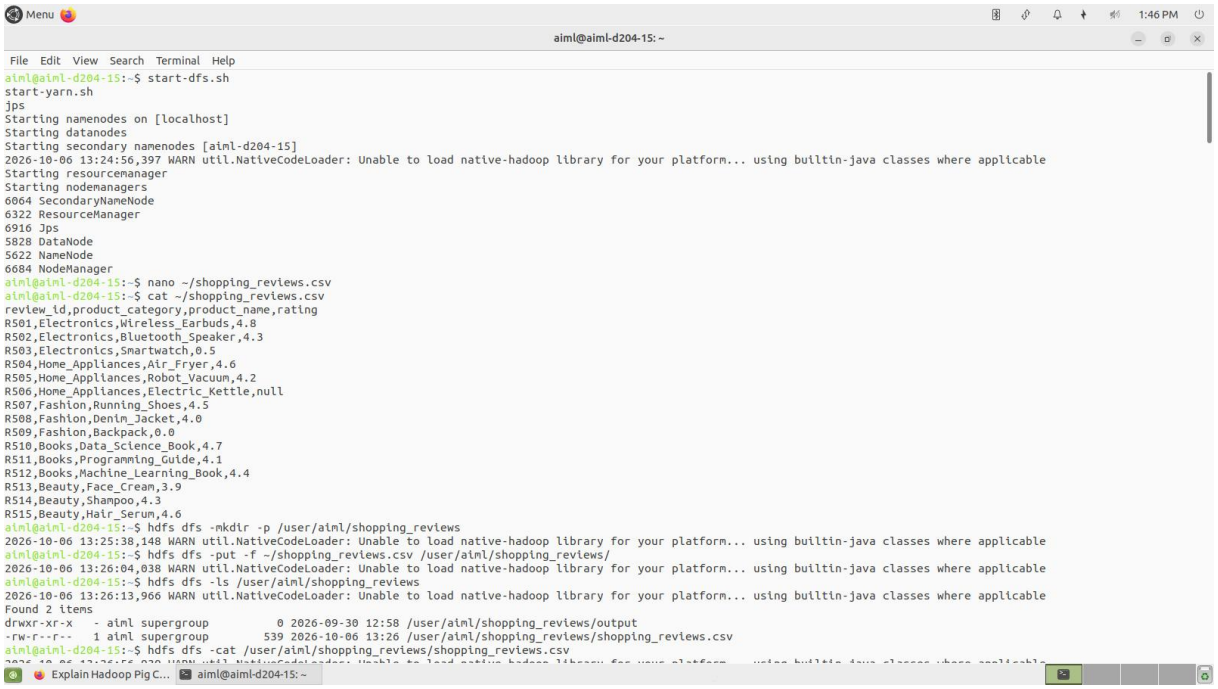
Screenshot 2 — Complete Pig Latin script and start of the execution command.



Screenshot 3 — Pig execution logs showing the script being submitted to the Hadoop/MapReduce environment.



Screenshot 4 — Execution completion/output verification area, including the HDFS output listing and category averages.



Screenshot 5 — Hadoop services started and the online-shopping review CSV dataset displayed in the terminal.